Explain Function

calc_vif()

Python

```
from statsmodels.stats.outliers_influence import
variance_inflation_factor

def calc_vif(X):

    #calculating VIF

    vif=pd.DataFrame()

    vif["variables"]=X.columns

    vif["VIF"]= [ variance_inflation_factor(X.values,i) for i
in range( X.shape[1])]

    return(vif)

#===================

calc_vif(dataset[['ssc_p', 'hsc_p', 'degree_p', 'etest_p',
'mba_p', 'salary']])

#===================
```

Ouput

	variables	VIF
0	ssc_p	78.168671
1	hsc_p	61.882196
2	degree_p	114.820554
3	etest_p	32.720365
4	mba_p	116.034378
5	salary	4.171783

```
#===================
```

This function is a standard way to quantify **multicollinearity** in the dataset. It uses the **Variance Inflation Factor (VIF)** to determine how much information provided by one variable is already "captured" or "duplicated" by the others.

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1. **decipher function line-by-line**

- **vif = pd.DataFrame()**: It initializes an empty table to store your results.
- **vif["variables"] = X.columns**: It takes the names of the features that was passed in (like ssc_p, hsc_p, etc.) and puts them in a column named "variables".
- **vif["VIF"] = [variance_inflation_factor(...) for i in range(X.shape[1])]**: This is a **list comprehension**. It loops through every column index (i) and calculates its specific VIF value using the statsmodels library.
- **return(vif)**: It outputs the final comparison table.

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2. The Logic: What is VIF actually doing?

To calculate the VIF for one variable (e.g., ssc_p), the function:

1. Temporarily treats ssc_p as a **dependent variable**.
2. Uses all the *other* variables (hsc_p, degree_p, etc.) as **independent variables**.
3. Calculates the **R^2 (Coefficient of Determination)** for that regression.
4. Plugs it into the VIF formula:

$$VIF = \frac{1}{1 - R^2}$$

- **If R^2 is high (Approaching 1)::** Indicates the variable is highly redundant, as it can be accurately predicted by other features. This results in a near-zero denominator, causing the **Big VIF value**(signalling severe multicollinearity)..
- **If R^2 is low (Approaching 0)**: Indicates the variable provides unique information. The denominator remains large, keeping the VIF near 1, which confirms the feature is independent.

The "Cause and Effect" Style

- **When $R^2 \approx 1$:** The feature is almost entirely explained by other columns. This creates a mathematical "bottleneck" in the VIF formula, leading to an **infinite or extremely high VIF value**.
- **When $R^2 \approx 0$:** The feature is unique. Since there is no overlap with other variables, the **VIF stays low**, indicating a healthy, independent feature.

Why this matters for the VIF formula:

In our code, we are calculating VIF using this relationship:

$$VIF = \frac{1}{1 - R^2}$$

- If $R^2 = 0.90$, then VIF = 1/ (1 - 0.90) = 10
  (The threshold where many analyst consider dropping a variable).
- If $R^2 = 0.99$, then VIF = 1/ (1 - 0.99) = 100
  (The value is **massively high**).

<u>Quick Tip :</u>

High R^2 ➔ Feature Redundancy

Low R^2 ➔ Feature Independence

3. inference on some Specific Output

In our result, most variables have **extremely high** VIF values:

Variable	VIF	Meaning
mba_p	**116.03**	Severe Multicollinearity
degree_p	**114.82**	Severe Multicollinearity
salary	**4.17**	Moderate / Acceptable

- Commonly, a VIF above **5 or 10** indicates that the variables are redundant. values (reaching 116) suggest that ssc_p, hsc_p, degree_p, and mba_p are so highly correlated that they are essentially telling the model the same thing over and over. This "noise" makes it hard for a regression model to know which variable actually influences the outcome.
- When VIF is this high, the standard practice is to **remove the variable with the highest VIF** and then run the function again. You'll likely see the other VIF values drop significantly!